子題一

```cpp
#include <iostream>

using namespace std;

const double pai=3.14159;

class Circle
{
protected:
    double radius;
public:
    Circle(double _r): radius(_r){}
    void print(){
        cout<<"Radius: "<<radius<<'\n';
    }
    double getPerimeter(){
        return (double)2*pai*radius;
    }
    double getArea(){
        return pai*radius*radius;
    }
};
```

```cpp
class Sphere : public Circle {
public:
    Sphere(double _r) : Circle(_r) {}
    double getSurface(){
        return (double)2*radius*getPerimeter();
    }
    double getVolume(){
        return (double)4/3*radius*getArea();
    }
};
int main(){
    // Instantiation and printing perimeter and area of a circle
    Circle cir(6.0);
    cout << "Circle: " << endl;
    cir.print();
    cout << "Perimeter: ";
    cout << cir.getPerimeter() << endl;
    cout << "Area: ";
    cout << cir.getArea() << endl << endl;
    // Instantiation and printing perimeter and area of a sphere
    Sphere sph(6.0);
    cout << "Sphere: " << endl;
    sph.print();
    cout << "Surface: ";
    cout << sph.getSurface() << endl;
    cout << "Volume: ";
    cout << sph.getVolume() << endl << endl;
    return 0;
}
```

```
Circle:
Radius: 6
Perimeter: 37.6991
Area: 113.097

Sphere:
Radius: 6
Surface: 452.389
Volume: 904.778

Press any key to continue . . .
```

子題二

```
#include<iostream>

using namespace std;

class Course
{
private:
    string name;
    int unit,score;
public:
    Course(string _n,int _u,int _s):name(_n),unit(_u),score(_s){}
    Course():name(""),unit(0),score(-1){}

    string getName(){
        return name;
    }
    int getUnit(){
        return unit;
    }
    int getScore(){
        return score;
    }
};

```

```cpp
class Student
{
private:
    string name;
    Course *courses;
public:
    Student(string _n):name(_n){courses=new Course[100];}
    ~Student(){delete[] courses;}

    void addCourse(string _n,int _u,int _s){
        for(int i=0;i<100;i++){
            if(courses[i].getScore()==-1){
                courses[i]=Course(_n,_u,_s);
                return;
            }
        }
    }
    void print(){
        cout<<"Student: "<<name<<'\n';
        double sumGP=0,sumCredit=0;
        for(int i=0;i<100;i++){
            if(courses[i].getScore()==-1){break;}
            int s=courses[i].getScore(),u=courses[i].getUnit(),gp;
            char g;

            if     (s>=90){g='A';gp=4;}
            else if(s>=80){g='B';gp=3;}
            else if(s>=70){g='C';gp=2;}
            else if(s>=60){g='D';gp=1;}
            else          {g='E';gp=0;}

            cout<<"Course: "<<courses[i].getName()<<" Credit: "<<u<<" Score: "<<s<<" Grade: "<<g<<'\n';
            sumGP+=gp*u; sumCredit+=u;
        }
        cout<<"GPA: "<<(sumCredit>0?sumGP/sumCredit:0.0)<<'\n';
        cout<<"-------------------"<<'\n';
    }
};
```

```cpp

int main(){
    Student student1("Henry");
    student1.addCourse("History", 3, 86);
    student1.addCourse("C++", 5, 70);
    student1.print();
    Student student2("Juan");
    student2.addCourse("Chem", 5, 60);
    student2.addCourse("Calc", 3, 55);
    student2.addCourse("Eng", 3, 90);
    student2.addCourse("PE", 1, 78);
    student2.print();
    return 0;
}
```

```
Student: Henry
Course: History Credit: 3 Score: 86 Grade: B
Course: C++ Credit: 5 Score: 70 Grade: C
GPA: 2.375
-------------------
Student: Juan
Course: Chem Credit: 5 Score: 60 Grade: D
Course: Calc Credit: 3 Score: 55 Grade: E
Course: Eng Credit: 3 Score: 90 Grade: A
Course: PE Credit: 1 Score: 78 Grade: C
GPA: 1.58333
-------------------
Press any key to continue . . .
```